Question	Answer	Marks	Guidance
1	**Defining the problem**		Annotate with P.
	Vary h and measure I **or** h is the independent variable and I is the dependent variable	**1**	This mark could be awarded having read the candidate's script all the way through. **Not** h or I constant.
	Keep f constant	**1**	**Allow** same for constant
	Methods of data collection		Annotate with M0
	Labelled diagram of workable circuit including: • reed switch and capacitor connected correctly • ammeter to measure I and • voltmeter to measure E	**1**	**Note** M2, M3, D7 **Allow** oscilloscope,  for voltmeter **Allow** use of $I = V / R$ explained with oscilloscope or voltmeter connected across resistor **Allow** $E = IR$ from separate circuit including ammeter
	Workable circuit showing signal generator or a.c. power supply connected to terminals F and G	**1**	**Allow**  for a.c. supply
	Diagram showing two supported (circular) plates either vertically or horizontally e.g. from stands or on the bench	**1**	**Allow** bench as one of the supports for horizontal method **Allow** use of (little) spacers to separate the plates for horizontal orientation
	Description of method to determine h, e.g. calipers	**1**	**Allow** micrometer with detail of measuring the total distance and subtracting width of plates **Not** rule(r)
	Method of Analysis		Annotate with ✓.
	Plot a graph of I against $\frac{1}{h}$ or equivalent e.g. lg I against lg h	**1**	**Note** drawn axes take priority.

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Question	Answer	Marks	Guidance
1	Relationship valid <u>if</u> a <u>straight</u> line is produced <u>passing through the origin</u> **or** relationship valid <u>if</u> a <u>straight</u> line is produced <u>with gradient of -1</u> for lg I against lg h	**1**	**Note** Must be consistent with drawn graph. **Ignore** 'data' / 'graph' / 'experiment' / 'results' / 'correlation' valid.
	$K = \frac{\text{gradient}}{AfE}$ for I against $\frac{1}{h}$ **or** $K = \frac{10^{y\text{–intercept}}}{AfE}$ for lg I against lg h	**1**	$K = \frac{1}{AfE \times \text{gradient}}$ for $\frac{1}{h}$ against I **Note** Must be consistent with drawn graph.
	Additional detail including safety considerations	**6**	Annotate with numbered tick
	D1 Precaution linked to <u>preventing plates falling/moving</u> e.g. stand(s) holding plates are clamped to the bench to prevent plates falling		**Allow** precaution linked to <u>hot resistor</u>, e.g. do not touch / wear gloves with <u>hot resistor</u> **Allow** precaution linked to shock from <u>plates</u> or <u>capacitor</u>, e.g. do not touch plates or switch off (d.c.) circuit when adjusting plates / h
	D2 Keep A and E constant		**Allow** same for constant **Allow** keep d.c. voltage supplied (to capacitor) for E
	D3 Use calipers / rule(r) to measure the diameter d of the plates **and** $A = \frac{\pi d^2}{4}$.		**Allow** $r = \frac{d}{2}$ and $A = \pi r^2$ **Not** micrometer
	D4 Use oscilloscope to determine f		**Allow** read off from signal generator / power supply
	D5 Method to determine f from oscilloscope, e.g. determine period T by multiplying the number of divisions in 1 cycle or length of 1 cycle by the time-base **and** $f = 1 / T$.		

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Question	Answer	Marks	Guidance
1	D6 Additional detail on use of oscilloscope to measure *T*, e.g. adjust time base for length of one cycle or measure *n* cycles and divide measured time by *n*		
	D7 Insulator on table / around supports		**Allow** from diagram
	D8 Use large values of *A* to <u>increase *I*</u> (to reduce percentage uncertainty of *I*)		**Allow** *E* or *f* for *A* **Allow** large values of radius or diameter for *A* **Ignore** increasing the p.d. / amplitude of the a.c. supply
	D9 Repeat measurement of *h* at different positions between the plates and determine a mean value for *h*		**Note** this is not dependent on M4
	D10 Description of method to keep the plates parallel and aligned (so that *h* is constant), e.g. use spacers of equal thickness **or** measure distances at different points on the plates to check is the same **or** use set squares on each side to check alignment **or** check plumb line touches both plates independently (for vertical alignment) **or** spirit level resting against both plates		**Note** explanation needed

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